

StartupShield AI

An AI-powered early-warning risk dashboard for startups & SaaS companies

The Idea

Startups usually don't lose customers or revenue suddenly — the warning signs show up early, but scattered across different places: a customer quietly using the product less, a support review turning negative, a strange dip in daily revenue, or growth slowing down. Because these signals live in different tools, founders often connect the dots only after the damage is done.

StartupShield AI solves this by combining four separate AI models into a single place, producing one easy-to-read **Risk Score (0–100)** for a company, along with the reason behind that score.

What It Does

Module	What it predicts	Technique
Churn Prediction	Which customers are likely to cancel	LightGBM / XGBoost / Random Forest
Sentiment Analysis	Whether reviews/tickets are positive, neutral, or negative	TF-IDF + Logistic Regression
Anomaly Detection	Unusual spikes/drops in revenue, users, or signups	Isolation Forest
Revenue Forecasting	Next 30 days' revenue, with a confidence range	Prophet / Holt-Winters (ETS)

All four outputs are then combined by a **Risk Aggregator** into one final score, with **SHAP**-based explainability showing exactly which factors drove that score.

Why It's Useful

- One dashboard instead of four separate tools/spreadsheets.
- Explainable — the score comes with a "why," not just a number.
- Every model is tested against a proper baseline, not just trained and assumed to work.

- Runs on open-source tools, no paid API or GPU needed.

Current Status

The four prediction modules (churn, sentiment, anomaly, forecasting) are built and tested — every one meets its target accuracy, beating a proper baseline. The final Risk Aggregator, explainability layer, and dashboard are the next steps.